

Manual for RiBESS+

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Introduction

This online application implements statistical methods for estimating the sample size, global (and group) sensitivity and probability of freedom from disease. It extends the features of the risk based estimate of system sensitivity tool (RiBESS). We refer the user to the according technical document A framework to substantiate absence of disease for the theory behind the implemented functions.

Installation

In this section we explain how to install the package on your computer. Alternatively it can be opened via <https://efsa.openanalytics.eu/app/ribess>, provided you have an account.

Prior to the installation, please make sure that the proxy is configured properly to access the internet from EFSA premises:

```
library(httr)
set_config(use_proxy(url="tmgproxy", port=8080, username="user",password="password"))
```

To run the application you first need to install the most recent version of the package “samplerator” in R. Then load the package and start the application with the runRibess() function.

```
library(samplerator)

runRibess()
```

Getting started

The the start up screen of the online application is shown below.

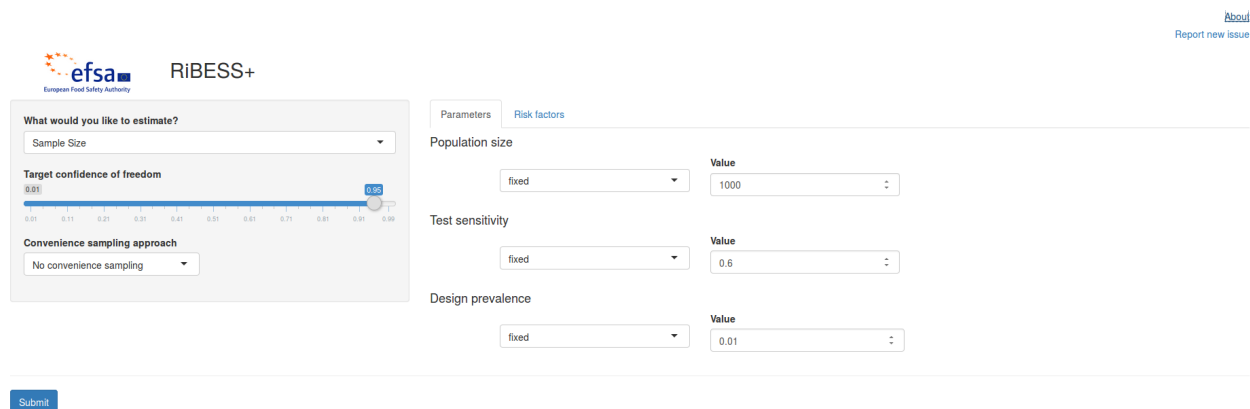


Figure 1: Start screen for the RiBESS+.

In the left panel there are four options for **What would you like to estimate?:**

- ‘Sample Size’: We aim to estimate the sample size for simple random sampling or a risk-based sampling design. This option should be chosen prior to the survey to know how many samples have to be collected to attain a certain target confidence of freedom.
- ‘Design Prevalence’: We aim to estimate the design prevalence for simple random sampling (risk-based sampling option not available). Estimation of this parameter can inform the researcher which design prevalence could be detected given the sample size and targeted confidence of freedom.
- ‘Global (and Group) Sensitivity’: We aim to estimate the global (and group) sensitivity for simple random sampling or a risk-based sampling design. This option should be chosen when the survey is over or ongoing as it estimates the sensitivity for each risk group and the global sensitivity when the sample size is provided.
- ‘Probability of Freedom’: We aim to estimate the probability of freedom from disease over time, taking into account previous surveillance results.

At the bottom there is a button **Submit** that has to be pushed to show the output. Every time the input values are changed, the submit button has to be pushed again to update the output.

In this manual we will give an overview of the available features for each subject of estimation. Various examples will then illustrate their use in practice.

About and Report new issue

The user is given a summary of the application's functionalities when clicking on the link "About" at the right top. New issues can be reported via "Report new issue". You are then redirected to the EFSA Model Manager website and invited to describe the issue.

Sample Size Estimation

The options 'Sample Size', 'Design Prevalence' and 'Global (and Group) Sensitivity' share the two tab panels at the right hand side.

As shown in Figure 1, the left panel displays two parameters:

- The **Target confidence of freedom** indicates the probability that is targeted for qualifying a statement on the presence/absence of a given disease.
- A **Convenience sampling approach** can be chosen out of: 'No convenience sampling', 'Convenience', 'Convenience with lower limit', 'Convenience with upper limit'. The former is the default method, the others are useful when it is known that it will be difficult to sample from some of the identified risk groups (see later).

At the right hand side there are two tabs: one for the 'Parameters' and one for the 'Risk factors'.

The 'Parameters' tab

This tab includes three general parameters:

- **Population size:** If the population size is limited, a value indicating its size should be provided here. If the population size is unlimited, this value can be ignored.
- **Test sensitivity:** The probability that the test used will return a positive result given that the sample is really positive.
- **Design prevalence:** The survey will be designed in order to obtain at least a positive test result when the prevalence of the disease will be at or above the defined value of the design prevalence.

Given a design prevalence of 1%, the target confidence of freedom thus expresses the probability of getting at least one positive test given that the infection is present at or above 1% of the subjects in the population.

The values for these parameters are **fixed** by default. However, in practice the exact values for population size and test sensitivity can be unknown. In that case, an appropriate random distribution can be assumed by changing fixed to **random - enter distribution quantiles** or

random - enter distribution parameters. If any of these options is chosen, further information is required in the input fields that pop up. In the left panel the **Number of random values** can be increased. A warning is shown when the number of random values is more than 1000, because this will seriously slow down calculations.

It is important to note that the formulae used for the sample size and sensitivity calculation assume a diagnostic test with 100% specificity.

If no risk groups can be identified in the population, the default settings in the Risk factors tab can be left unchanged. We will illustrate this with three examples.

Example without risk groups, fixed input

We use the default fixed values: A population size of 1000, test sensitivity of 0.6 and design prevalence 0.01. We estimate the sample size for a targeted confidence of freedom of 95%. The output is given in Figure 2

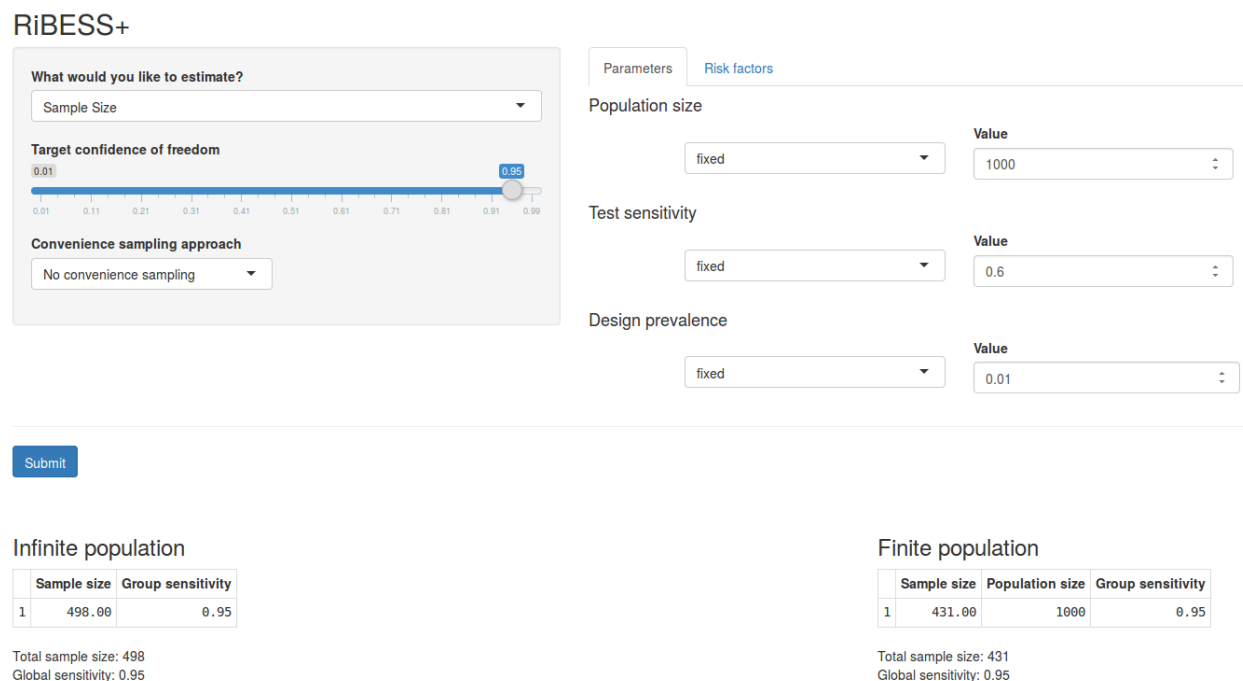


Figure 2: Output for example without risk groups, fixed input.

Two tables are shown: one for an **Infinite population** and one for a **Finite population**. The population can be considered infinite if the individual probability of being positive does not change along the sampling exercise. Then a Binomial distribution is used for the estimation and the input value for population size is ignored. Notice that when the input value for population size is very large, both tables will hold similar results.

Each table reports the estimated sample size and group sensitivity. The latter equals the global sensitivity when no risk groups are defined. For the finite population case also the population size for each group is reported.

Example without risk groups, random - enter distribution quantiles

For example, the scientist may be uncertain about the exact value of the test sensitivity, but he/she estimates that approximately 25% of the test sensitivity values will be below 60% and 25% of the test sensitivity values will exceed 70%. According to this information, the program will automatically select the distribution (out of normal, beta and gamma) that results in the largest log-likelihood value. Then, for the given parameter estimates, 100 random values for the test sensitivity will be generated and a density curve will be plotted at the right. The sample size will be estimated for each of the generated values of the test sensitivity.

You can click on a single cell in the Quantile - Value matrix to enter a numeric value. To overwrite the content of a cell, select the current input value and type the new value. The content of a cell can not be deleted. We can increase/decrease the number of quantile values by pushing the +/- button, and obtain Figure 3. Notice that the quantiles should be ordered from small to large, otherwise an error message is shown.

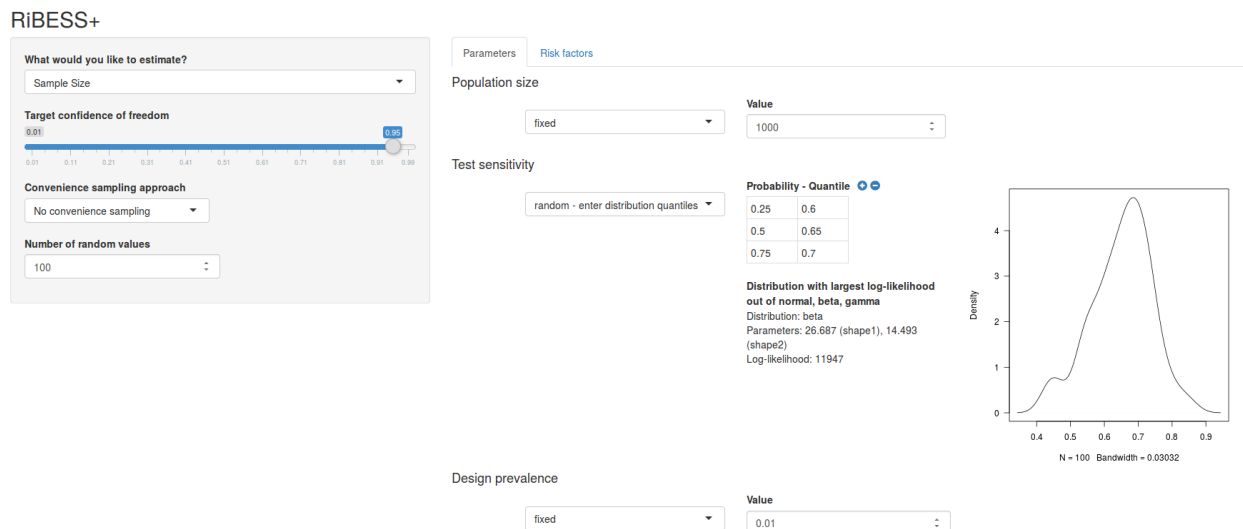


Figure 3: Test sensitivity: Random - enter distribution quantiles.

We define test sensitivity as in the figure above. For the other parameters we use the default fixed values: A population size of 1000, design prevalence 0.01 and 100 random values. We estimate the sample size for a targeted confidence of freedom of 95%. The output is given in Figure 4.

On top it is stated that results are summarized according to a **parameter**, which is either 'Sample size', 'Group sensitivity' or 'Population size'. An approximation of the cumulative distribution of this parameter is plotted at the bottom, while exact values are given in the table. This is an interactive plot in the sense that the cursor can be moved over the plotted line to show the corresponding approximate values. The input field **Show table for percentile** indicates for which percentile of the chosen parameter the table is shown, by default this is the median of sample size. Alternatively it can be changed to 0.25 for the first quantile or any other value. The tables and figures can be downloaded and saved as respectively a .csv or .html file.

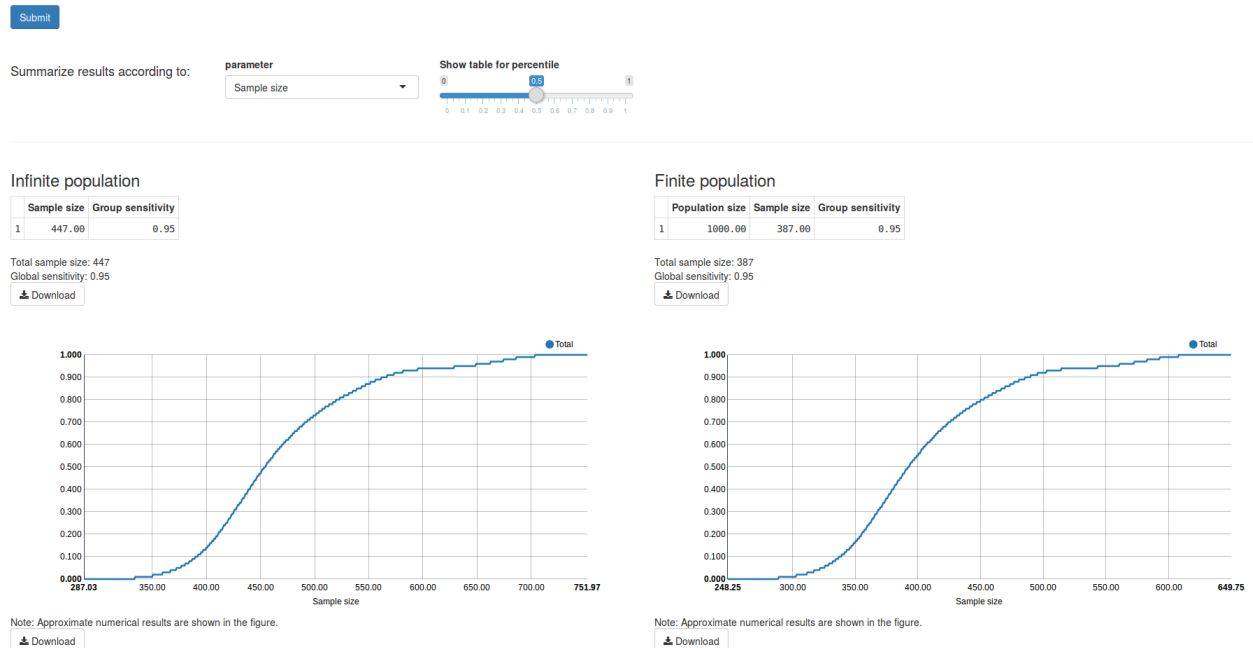


Figure 4: Output for test sensitivity: Random - enter distribution quantiles.

Example without risk groups, random - enter distribution parameters

Alternatively, the test sensitivity can be assumed to follow a uniform distribution with known minimum 0.5 and maximum 0.8, see Figure 5. The assumed **distribution** for this parameter can be selected from ‘uniform’, ‘normal’, ‘beta’ or ‘gamma’.

Test sensitivity



Figure 5: Random - enter distribution parameters.

The output structure is as discussed for entering distribution quantiles.

The ‘Risk factors’ tab

This tab allows the user to define the relevant risk groups that can be identified in the population.

There are three options for **Enter as data frame**:

- If “none” is selected, the shown input fields ask to enter the data per individual risk factor
- If “Proportion” is selected, the proportions should be entered per risk factor combination in the data frame shown at the bottom.
- if “Relative risk and Proportion” is selected, the relative risks and proportions should be entered per risk factor combination in the data frame at the bottom.

We now advise to follow these steps:

- Increase the **Number of Risk factors** to the required number. For 2 risk factors the result is shown in Figure 6.
- Optionally, the name for each risk factor can be changed in the field **Risk Factor**.
- Set the number of levels per risk factor with **# levels**. Notice that the number of levels per risk factor is at least 2, otherwise it is not relevant to define the risk factor.
- Per risk factor, you can optionally change the name for each level in **Level name**.
- The **Relative risk** defines the risk of disease introduction compared to some reference group, which has relative risk 1.
- The **Proportion** defines the relative proportion of the population that is expected to be allocated in each group.

Values for the relative risks and proportions may be known from previous studies or originate from expert knowledge, they are ‘fixed’ by default. Alternatively, they are ‘random’ when for example based on the output from a fitted model.

The first two examples assume fixed input values, while the last two examples illustrate the use of random input values.

Example with risk groups, fixed input - Enter as data frame: none

Consider an example where two risk factors were identified with 3 levels each, see Figure 7.

In this case, a relative risk of 3 for type Stray states that the probability of disease introduction is 3 times larger in this group compared to the Domestic group. Notice that exactly one reference group per risk factor can be defined, otherwise levels with relative risk 1 could be combined per risk factor. An error message will warn the user for incorrect input values.

The expected proportion per risk factor should sum to 1. Therefore, it is not necessary to enter all proportions, they can be completed by pushing the button: **Complete risk proportions**. By doing so, missing values are replaced by one and the same value such that the sum of all proportions is 1. So, for the example we can enter the values in Figure 8 and then push the button.

For the other parameters we use the default fixed values: A population size of 1000, test sensitivity of 0.6 and design prevalence 0.01, we can estimate the sample size for a targeted confidence of freedom of 95%. The output is given in Figure 9.

Parameters
Risk factors

Enter as data frame

<none>

Number of Risk factors

0220

Complete risk proportions

Relative risk:

fixed

Proportion:

fixed

Risk Factor

risk factor 1

levels

210

Level name

a

Value

1

Value

Level name

b

Value

1

Value

Risk Factor

risk factor 2

levels

210

Level name

a

Value

1

Value

Level name

b

Value

1

Value

Figure 6: Increase the number of risk factors.

Parameters
Risk factors

Enter as data frame
<none>

Number of Risk factors
0 2 20

Complete risk proportions

Relative risk:
fixed

Proportion:
fixed

Risk Factor
Type

levels
2 3 10

Level name
Value
Value

Stray
3
0.1

Level name
Value
Value

Hunting
2
0.2

Level name
Value
Value

Domestic
1
0.7

Risk Factor
Age

levels
2 3 10

Level name
Value
Value

Old
3
0.4

Level name
Value
Value

Adult
2
0.4

Level name
Value
Value

Young
1
0.2

Figure 7: Entering risk factor information per factor.

Proportion:

fixed

Value

0.1

Value

0.2

Value

Value

Value

Value

0.2

Figure 8: Auto-complete risk proportions.

Infinite population

	Type	Age	Sample size	Group sensitivity
1	Stray	Old	19.00	0.28
2	Stray	Adult	29.00	0.28
3	Stray	Young	57.00	0.28
4	Hunting	Old	29.00	0.28
5	Hunting	Adult	43.00	0.28
6	Hunting	Young	86.00	0.28
7	Domestic	Old	57.00	0.28
8	Domestic	Adult	86.00	0.28
9	Domestic	Young	171.00	0.28

Total sample size: 577
Global sensitivity: 0.95

Finite population

	Type	Age	Population size	Sample size	Group sensitivity
1	Stray	Old	40.00	19.00	0.32
2	Stray	Adult	40.00	26.00	0.32
3	Stray	Young	20.00	20.00	0.16
4	Hunting	Old	80.00	29.00	0.32
5	Hunting	Adult	80.00	41.00	0.32
6	Hunting	Young	40.00	40.00	0.21
7	Domestic	Old	279.00	60.00	0.31
8	Domestic	Adult	279.00	86.00	0.31
9	Domestic	Young	139.00	131.00	0.31

Total sample size: 452
Global sensitivity: 0.95

Figure 9: Output for example with risk groups, fixed input.

Example with risk groups, fixed input - Enter as data frame: Relative risk and Proportion

We choose to enter both relative risk and proportion as data frame. The risk factor and level names are automatically copied into a matrix. Please do not change the names in the matrix, but use the input fields on top.

The button Complete risk proportions follows the same principles as before. The matrix corresponding to the previous example is given Figure 10.

Type	Age	Relative risk	Proportion
Stray	Old	9.00	0.04
Stray	Adult	6.00	0.04
Stray	Young	3.00	0.02
Hunting	Old	6.00	0.08
Hunting	Adult	4.00	0.08
Hunting	Young	2.00	0.04
Domestic	Old	3.00	0.28
Domestic	Adult	2.00	0.28
Domestic	Young	1.00	0.14

Figure 10: Entering risk factor information as data frame.

Example with risk groups, random input - Enter as data frame: none

When choosing ‘random’ instead of ‘fixed’, for each level a **Distribution** field pops up showing ‘reference’ by default. For the relative risk this implies a value 1, for the proportion it implies a value such that the sum per risk factor is 1. The distribution can be changed to ‘normal’, the user is then asked to provide parameter values for the mean and standard deviation. In the plot right to these parameter values, the randomly generated values are respectively shown on the *exponential scale* for the relative risks and on the *expit scale* for the proportions. A plot for the reference proportions is shown when pushing the button complete risk proportions. Again, the **Number of random values** can be increased.

We show a simple example in Figures 11 and 12, for 2 risk factors with each 2 levels.

For the other parameters we use the default fixed values: A population size of 1000, test sensitivity of 0.6 and design prevalence 0.01, we can estimate the sample size for a targeted confidence of freedom of 95%. The output is given in Figure 13.

The interactive plot shows one line per risk group and the line with label ‘Total’ represents the total sample size. The risk groups are numbered according the output in the tables. For example, risk group 1 concerns Type stray and Age old. We can hide a line by clicking once on the legend name, we can show a single line by double-clicking on the legend name.

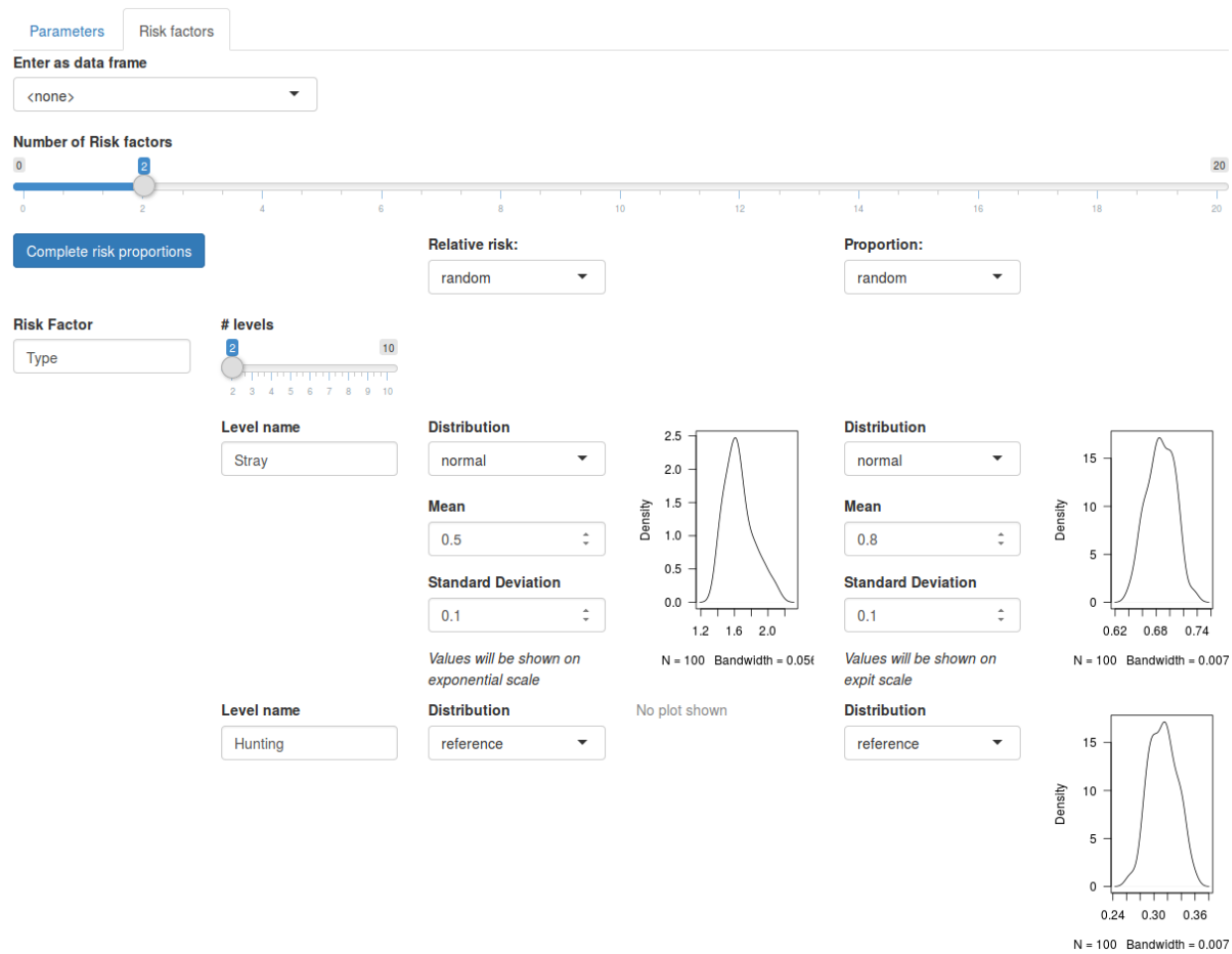


Figure 11: Entering random relative risk and proportion for risk factor Type.

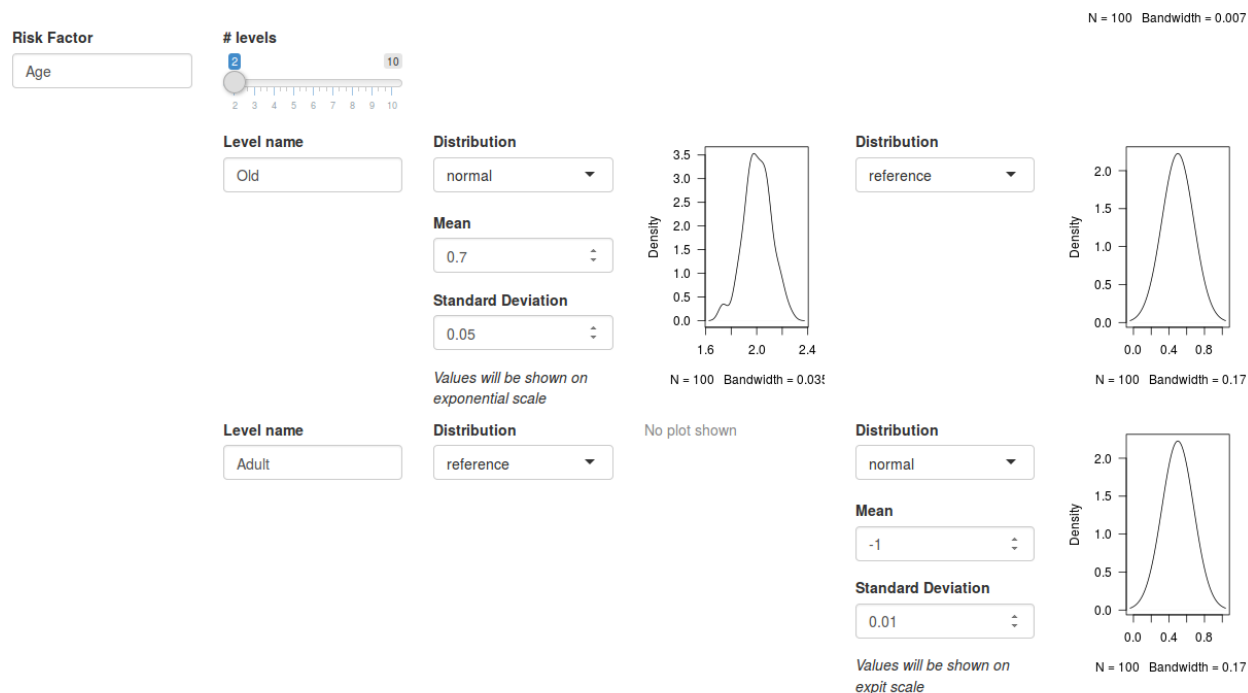


Figure 12: Entering random relative risk and proportion for risk factor Age.

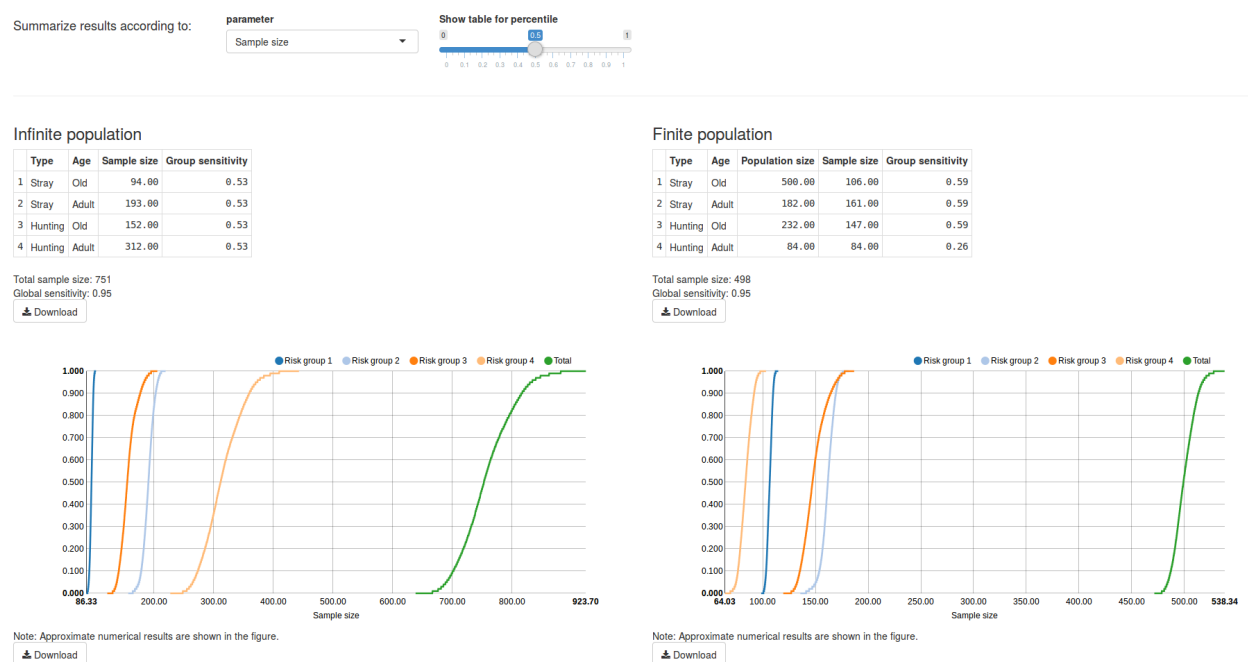


Figure 13: Output for example with random relative risks and proportions.

Example with risk groups, random input - Enter as data frame: Proportion

We choose to enter the proportion as data frame. The risk factor and level names are automatically copied into a matrix. Please do not change the names in the matrix, but use the input fields on top.

The button Complete risk proportions follows the same principles as before. The random relative risks and proportion matrix for this example are given in Figures 14 and 15.

Parameters Risk factors

Enter as data frame

Proportion

Number of Risk factors

0 2 20

Complete risk proportions

Relative risk: random

Proportion: fixed

Risk Factor Type

levels 2 10

Level name Stray

Distribution normal

Mean 0.5

Standard Deviation 0.1

Values will be shown on exponential scale

Level name Hunting

Distribution reference

Density

2.5 2.0 1.5 1.0 0.5 0.0

1.2 1.6 2.0

N = 100 Bandwidth = 0.05

No plot shown

Figure 14: Entering random relative risk for risk factor Type.

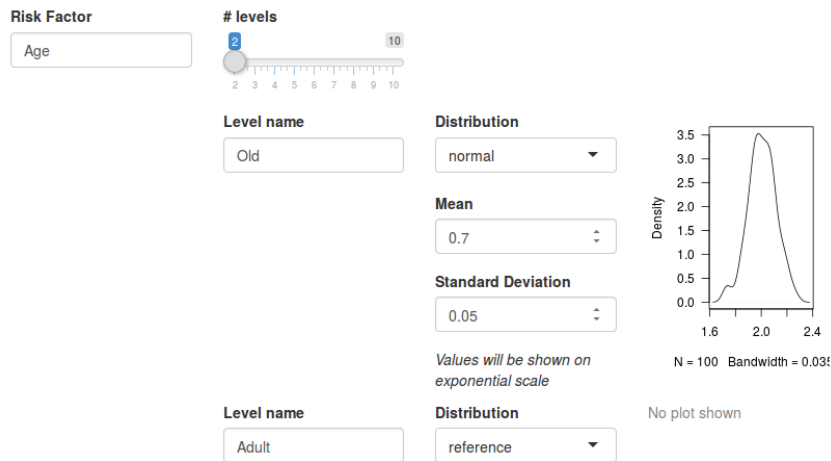
The corresponding results are given in Figure 16

Convenience sampling

In case it is known it will be difficult to sample from some of the identified risk groups in an epidemiologically relevant area, it is possible to change the **Convenience sampling approach**. We will illustrate each of the options with an example, except for 'No convenience' which use has been shown in the previous examples.

Example for 'Convenience'

Consider an example in which 4 risk groups cannot be sampled (Stray & Old, Stray & Adult, Hunting & Young and Domestic & Adult). From 2 risk groups, the user wants to minimize the sampling pressure (Hunting & Old and Domestic & Old): these can be defined as 'least convenient'



Type	Age	Proportion
Stray	Old	0.17
Stray	Adult	0.17
Hunting	Old	0.33
Hunting	Adult	0.33

Figure 15: Entering random relative risk for risk factor Age and fixed proportions.



Figure 16: Output for example with random relative risks.

risk groups with value 1. For the remaining 3 risk groups, i.e. Stray & Young, Hunting & Adult and Domestic & Young, it is planned to be able to sample respectively twice, 4 times and 3 times the amount of samples needed in the least convenient risk groups. We can enter this information as follows in the convenience matrix, see Figure 17.

RiBESS+

What would you like to estimate?

Sample Size

Target confidence of freedom

0.95

Convenience sampling approach

Convenience

The number sampled per group is proportional to the convenience value

Type	Age	Convenience
Stray	Old	0.00
Stray	Adult	0.00
Stray	Young	2.00
Hunting	Old	1.00
Hunting	Adult	4.00
Hunting	Young	0.00
Domestic	Old	1.00
Domestic	Adult	0.00
Domestic	Young	3.00

Parameters

Risk factors

Enter as data frame

<none>

Number of Risk factors

2

Complete risk proportions

Risk Factor

Type

levels

3

Level name

Stray

Value

3

Level name

Hunting

Value

2

Level name

Domestic

Value

1

Relative risk:

fixed

Proportion:

fixed

Value

0.1

Value

0.2

Value

0.7

Risk Factor

Age

levels

3

Level name

Old

Value

3

Level name

Adult

Value

2

Level name

Young

Value

1

Value

0.4

Value

0.4

Value

0.2

Figure 17: Entering data for ‘Convenience’ sampling.

For the other parameters, we use the default fixed values: A total population size of 1000, test sensitivity of 0.6 and design prevalence 0.01, we can estimate the sample size for a targeted confidence of freedom of 95%. The output is given in Figure 18. Notice that no group sensitivity value for Stray and Young can be calculated, because sample size is much larger than population size for that group.

Infinite population

No results, unless 'No convenience sampling'

Finite population

	Type	Age	Population size	Sample size	Group sensitivity
1	Stray	Old	40.00	0.00	0.00
2	Stray	Adult	40.00	0.00	0.00
3	Stray	Young	20.00	20.00	0.16
4	Hunting	Old	80.00	66.00	0.65
5	Hunting	Adult	80.00	80.00	0.61
6	Hunting	Young	40.00	0.00	0.00
7	Domestic	Old	279.00	66.00	0.34
8	Domestic	Adult	279.00	0.00	0.00
9	Domestic	Young	139.00	139.00	0.34

Total sample size: 371
Global sensitivity: 0.95

Download

Figure 18: Output for example with ‘Convenience’ sampling.

Example for ‘Convenience with lower limit’

When choosing ‘Convenience with lower limit’, the user is asked to provide a value for **Limit**. The latter defines the number of samples that will be taken for the risk group with convenience 1, i.e. the least convenient risk group. We set the value to 40 and get the warning that population size is too small if set to 1000. For illustrative purpose, we therefore use set population size to 2000 and obtain the result in Figure 19.

Infinite population

No results, unless ‘No convenience sampling’

Finite population

	Type	Age	Population size	Sample size	Group sensitivity
1	Stray	Old	80.00	0.00	0.00
2	Stray	Adult	80.00	0.00	0.00
3	Stray	Young	40.00	40.00	0.30
4	Hunting	Old	160.00	40.00	0.40
5	Hunting	Adult	160.00	150.00	0.82
6	Hunting	Young	80.00	0.00	0.00
7	Domestic	Old	559.00	40.00	0.21
8	Domestic	Adult	559.00	0.00	0.00
9	Domestic	Young	279.00	100.00	0.20

Total sample size: 370

Global sensitivity: 0.95

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Figure 19: Output for example with ‘Convenience with lower limit’ sampling.

So the number of samples from the least convenient risk groups is automatically 40, and the remaining needed samples are taken from the other risk groups with convenience 2, 3 and 4.

Example for ‘Convenience with upper limit’

When choosing ‘Convenience with upper limit’, the user is asked to provide a value for **Limit**. The latter defines the maximum number of samples that will be taken for the risk group with convenience 1, i.e. the least convenient risk group. We set the value to 20 and population size to 2000, and we obtain the result in Figure 20

Infinite population

No results, unless ‘No convenience sampling’

Finite population

	Type	Age	Population size	Sample size	Group sensitivity
1	Stray	Old	80.00	0.00	0.00
2	Stray	Adult	80.00	0.00	0.00
3	Stray	Young	40.00	40.00	0.30
4	Hunting	Old	160.00	20.00	0.22
5	Hunting	Adult	160.00	160.00	0.85
6	Hunting	Young	80.00	0.00	0.00
7	Domestic	Old	559.00	20.00	0.11
8	Domestic	Adult	559.00	0.00	0.00
9	Domestic	Young	279.00	160.00	0.32

Total sample size: 400

Global sensitivity: 0.95

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Figure 20: Output for example with ‘Convenience with upper limit’ sampling.

So the number of samples from the least convenient risk groups is 20, and the remaining needed samples are taken from the other risk groups with convenience 2, 3 and 4.

Design Prevalence Estimation

In order to estimate the design prevalence, we need to enter the **Sample size** instead of design prevalence in the parameters tab-panel. In the risk factors tab-panel it is indicated that “Risk factors analysis is not available when estimating Design Prevalence”. In general, the procedure for estimating the design prevalence is very similar to the one for estimating sample size, as shown previously.

For example, we can enter values as shown in Figure 21 resulting in the output in Figure 22.

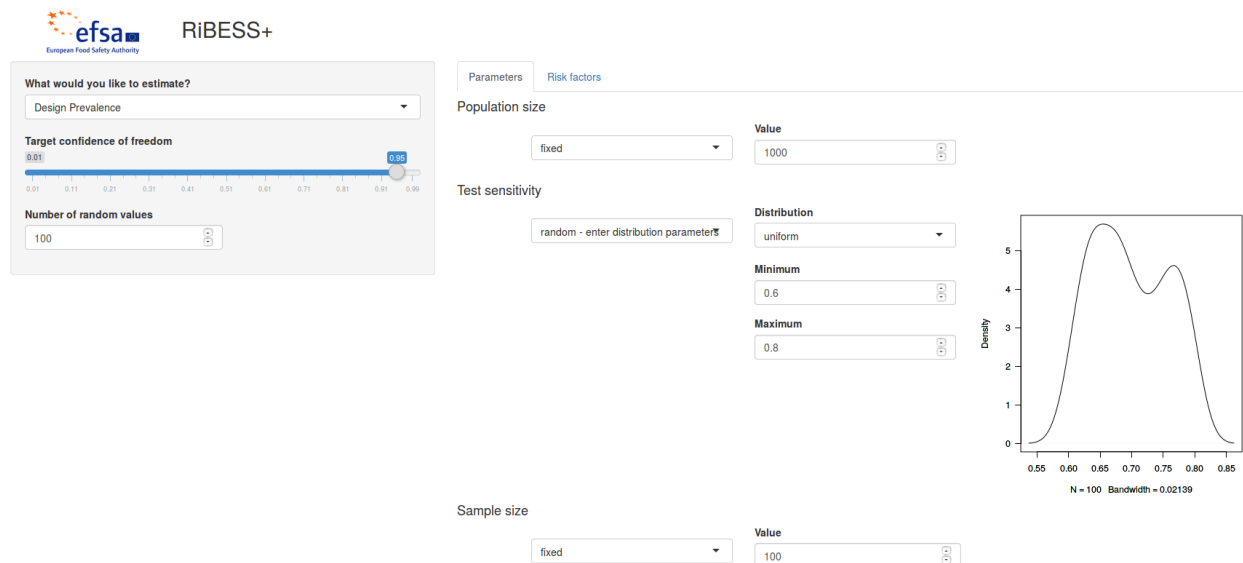


Figure 21: Input for design prevalence estimation.

Global (and Group) Sensitivity Estimation

When the aim is to estimate the global (and group) sensitivity, we need to enter the **Sample Size** (for each risk group) in the displayed matrix. You can click on a single cell to enter the Sample Size. To overwrite the content of a cell, select the current input value and type the new value.

For example, consider the sample sizes as shown in Figure 23. We assume that the test sensitivity follows a uniform distribution with minimum 0.5 and maximum 0.7. The other parameters have the default values. We can then plot the group sensitivity per risk group, as shown in Figure 24.

Probability of Freedom from Disease Estimation

In this section we illustrate how we can take into account previous surveillance results to estimate the probability of freedom from disease. Indeed, the confidence that a member state “A” is free from a given infection (i.e. design prevalence $< 1\%$) would be higher if previous surveillance activities had been carried out without finding infected animals when compared to a member state “B”, which

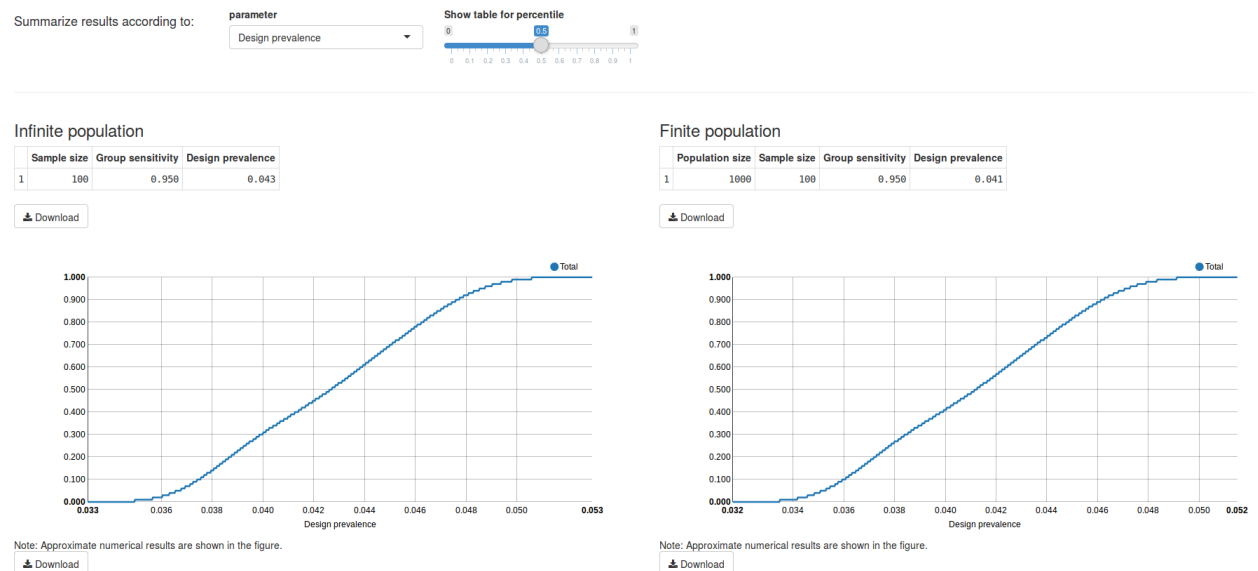


Figure 22: Output for design prevalence estimation.

RIBESS+

What would you like to estimate?
Global (and Group) Sensitivity

Type	Age	Sample Size
Stray	Old	20.00
Stray	Adult	20.00
Stray	Young	30.00
Hunting	Old	30.00
Hunting	Adult	40.00
Hunting	Young	50.00
Domestic	Old	60.00
Domestic	Adult	70.00
Domestic	Young	80.00

Enter as data frame
<none>

Number of Risk factors
0

Complete risk proportions

Risk Factor
Type

levels
2

Level name	Value
Stray	3
Hunting	2
Domestic	1

Relative risk:
fixed

Proportion:
fixed

Risk Factor
Age

levels
3

Level name	Value
Old	3
Adult	2
Young	1

Value
0.1
0.2
0.7

Value
0.4
0.4
0.2

Figure 23: Enter sample size for global (and group) sensitivity estimation.

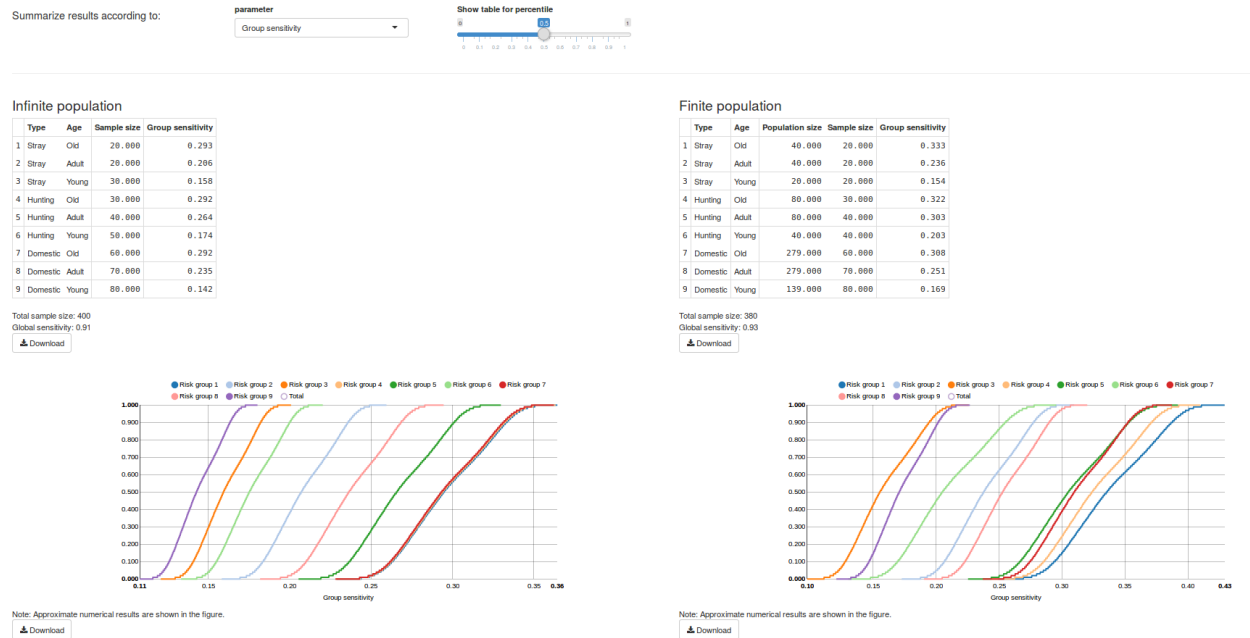


Figure 24: Output for global (and group) sensitivity estimation with random test sensitivity.

is implementing a survey for the first time. To address this issue, we estimate the Probability of Freedom from disease.

To estimate the probability of freedom from disease we need to provide values for the following parameters:

- The **Target confidence of freedom** indicates the probability that is targeted for qualifying a statement on the presence/absence of a given disease.
- The **Global sensitivity in time period 1** indicates the probability of getting at least one positive test given that the infection is present at or above the prevalence in the target population during the first time period. This value can either be entered by the user or copied from the last results for group sensitivity by ticking **Copy from Estimation Results**.
- The **Prior probability of freedom (for time period 1)** is the probability of the infection being absent (i.e. smaller than the design prevalence) given that all samples tested negative (at the start of the first time period). This parameter is fixed by default, when random it can be defined by entering distribution quantiles or parameters.
- The user is first asked how many time periods should be considered in **# periods**. For each time period it is then asked to enter the **Probability of disease introduction**. For example for time period 1 it indicates the probability that the disease is introduced between start of the study and the start of the time period 2. Each of these probabilities can be entered as fixed or random parameter.

Example for probability of freedom, fixed input

For the default values of the input parameters, we obtain the output in Figure 25.

RiBESS+

What would you like to estimate?

Probability of Freedom from Disease

Target confidence of freedom

0.01 0.11 0.21 0.31 0.41 0.51 0.61 0.71 0.81 0.91 0.99

0.95

Global sensitivity in time period 1

0.95

☐ Copy from Estimation Results

Prior probability of freedom (for time period 1)

fixed

Value

0.5

Probability of disease introduction

periods

2 3 4 5 6 7 8 9 10

3

Time period 1

fixed

Value

0.2

Time period 2

fixed

Value

0.2

Submit

	Prior probability of freedom	Unadjusted prob. of freedom	Global sensitivity	Prob. of disease introduction	Adjusted prob. of freedom
Time period 1	0.50	0.95	0.95	0.20	0.76
Time period 2	0.76	0.95	0.84	0.20	0.76

Figure 25: Output for probability of freedom, fixed input.

A table is reported with for each time period:

- The **Prior probability of freedom** indicates the probability of freedom from disease at the start of that time period. Notice that for the first time period, it is a copy of the user input.
- The **Unadjusted probability of freedom** is the probability of freedom from disease incorporating historical evidence of freedom from previous surveillance activities, but not accounting for the probability of disease introduction during that time period.
- **Global sensitivity** indicates the sensitivity during each time period. Notice that for the first time period, it is a copy of the user input.
- **Probability of disease introduction**: copy from user input
- The **Adjusted probability of freedom** is the probability of freedom from disease adjusted for the fact that the disease is introduced during this time period with the corresponding value for probability of disease introduction.

Example for probability of freedom, copy Global sensitivity in time period 1

We can easily use Group sensitivity values from the last estimation results by ticking the box for copying. We can choose the risk group from the drop down menu **for selected risk group**. An example is given in Figure 26.

Example for probability of freedom, random input

We assume that the probability of disease introduction for the first time period follows a uniform distribution with minimum 0.1 and maximum 0.3. The other parameters are left unchanged, so

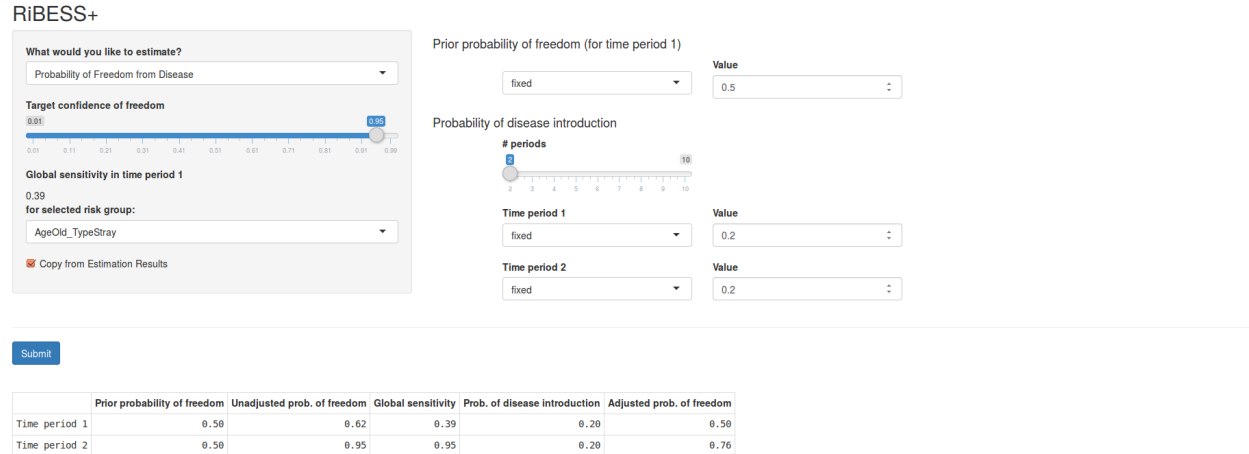


Figure 26: Output for probability of freedom, fixed input.

that the output is given in Figure 27.

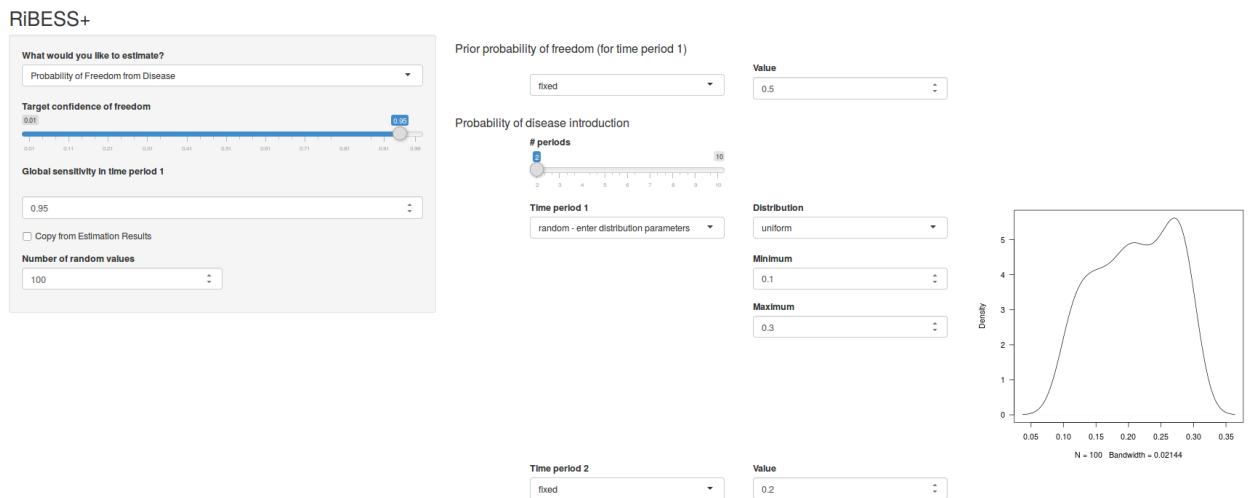


Figure 27: Random input for estimating probability of freedom.

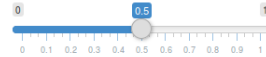
As before for random input values, the results are summarized according to one of the variables in the table and the table is shown for the median by default (see Figure 28). In the figure, a vertical line is shown for parameters that have constant values.

Summarize results according to:

parameter

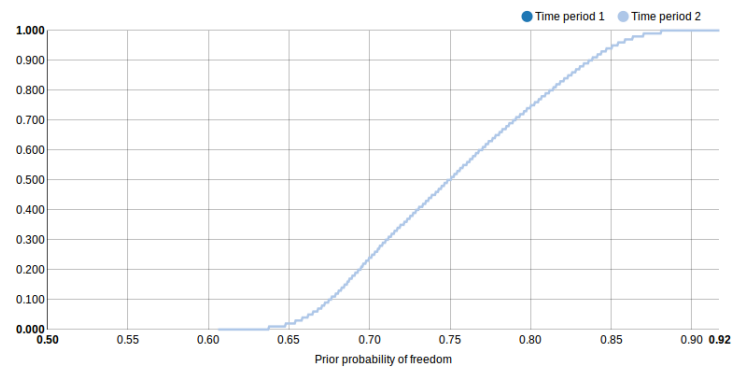
Prior probability of freedom

Show table for percentile



	Prior probability of freedom	Unadjusted prob. of freedom	Global sensitivity	Prob. of disease introduction	Adjusted prob. of freedom
Time period 1	0.500	0.952	0.950	0.213	0.749
Time period 2	0.749	0.952	0.850	0.200	0.762

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Note: Approximate numerical results are shown in the figure.

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Figure 28: Output for probability of freedom, random input.